

Study workflow

Medullary thyroid carcinoma cohort
n=4,387

Patients eligible for analysis
n=3,606

Excluded from analysis:
Age < 18 years, n=138
Race other than White or Black, n=301
Tumor size missing or <= 0, n=342

**Data
preparation**

1. Check variables, missingness and outcome definition
2. Separate development and validation data before preprocessing
3. Estimate transformations using development data only

**Model
selection**

1. Compare candidate algorithms with cross-validation
2. Choose tuning parameters within the development data

**Model
development**

1. Fit the selected model with the chosen parameters

**Model
validation**

1. Evaluate discrimination and calibration on held-out data
2. Report performance and uncertainty

**Model
explanation**

1. Summarize feature contributions with SHAP

Patient selection